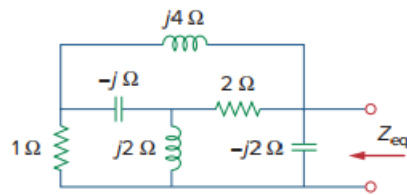


Problem

Obtain the equivalent impedance of the circuit in Fig. 9.78.

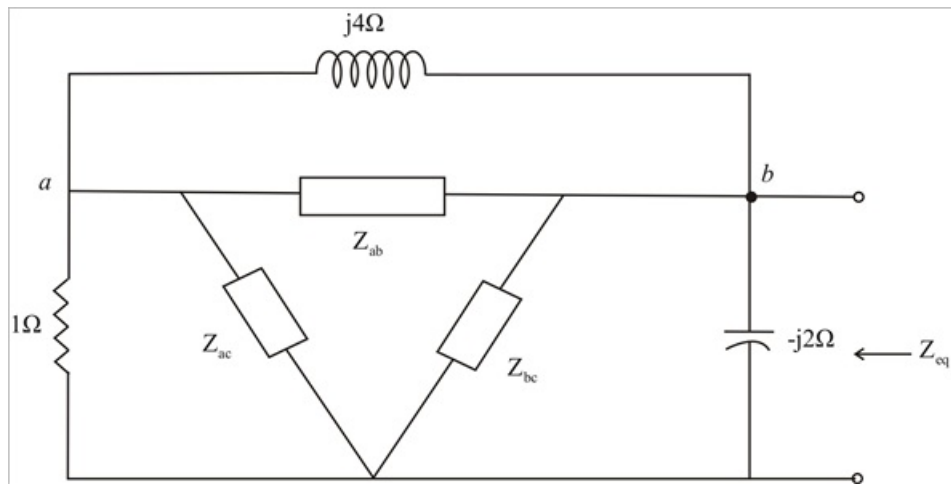
Figure 9.78:



Step-by-step solution

Step 1 of 3

We apply a WYE – to – delta transformation.



Step 2 of 3

$$Z_{ab} = \frac{2 - j2 + j4}{j2} = \frac{2 + j2}{j2} = 1 - j$$

$$Z_{ac} = \frac{2 + j2}{2} = 1 + j$$

$$Z_{bc} = \frac{2 + j2}{-j} = -2 + j2$$

Step 3 of 3

$$j4 \parallel Z_{ab} = j4 \parallel (1-j) = \frac{(j4)(1-j)}{1+j3} = 1.6 - j0.8$$

$$1 \parallel Z_{ac} = 1 \parallel (1+j) = \frac{(1)(1+j)}{2+j} = 0.6 + j0.2$$

$$j4 \parallel Z_{ab} + 1 \parallel Z_{ac} = 2.2 - j0.6$$

$$\frac{1}{Z_{eq}} = \frac{1}{-j2} + \frac{1}{-2+j2} + \frac{1}{2.2-j0.6}$$

$$= j0.5 - 0.25 - j0.25 + 0.4231 + j0.1154$$

$$= 0.173 + j0.3654 = 0.4043 \angle 64.66^\circ$$

$$Z_{eq} = 2.473 - j64.66^\circ \Omega = 1.058 - j2.235 \Omega .$$